

Chapter 14

Description of Motion (Part Three)

Terms List

(4) Average Acceleration

(5) Instantaneous Acceleration

(5) Instantaneous Acceleration Vector

14.7 Acceleration

- In everyday language, we say that a car accelerates when its speed increases and it decelerates when its speed decreases.
- In physics, acceleration is a quantity used to describe the variations of the velocity with respect to time.

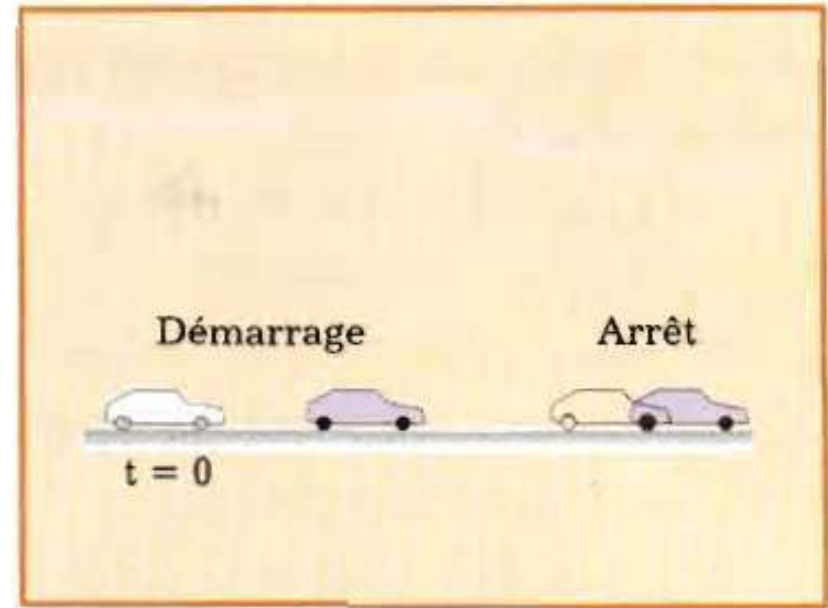
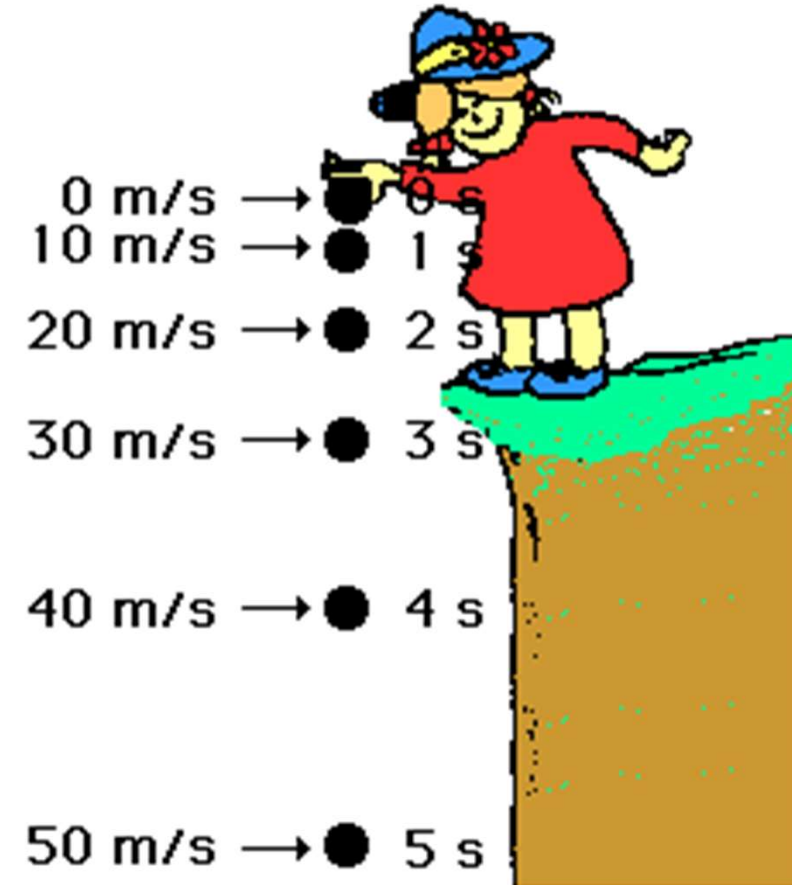


Figure 14.12. Démarrage et arrêt d'une voiture.

Example of Acceleration

- The initial speed is zero.
- The speed of the falling object increases by 10 m/s every second.
- Thus the **average acceleration** of the falling body is 10 m/s/s or 10 m/s².



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graph TD; A[Acceleration] --> B[Average]; A --> C[Instantaneous];
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Acceleration

Average

Instantaneous

(5) Average Acceleration

- V_1 instantaneous speed at instant t_1 .
- V_2 instantaneous speed at instant t_2 .

The average acceleration a_a of the moving object between the instants t_1 et t_2 is:

$$\text{Average acceleration} = \frac{\text{change in speed}}{\text{time taken}}$$

$$a_{av} = \frac{v_2 - v_1}{t_2 - t_1}$$

$$a_{av} = \frac{\Delta v}{\Delta t}$$

In the SI system, the acceleration is measured in m/s^2 ou m.s^{-2} .

(5) Average Acceleration



$$a_a = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i}$$

Evaluation

Acceleration = ?

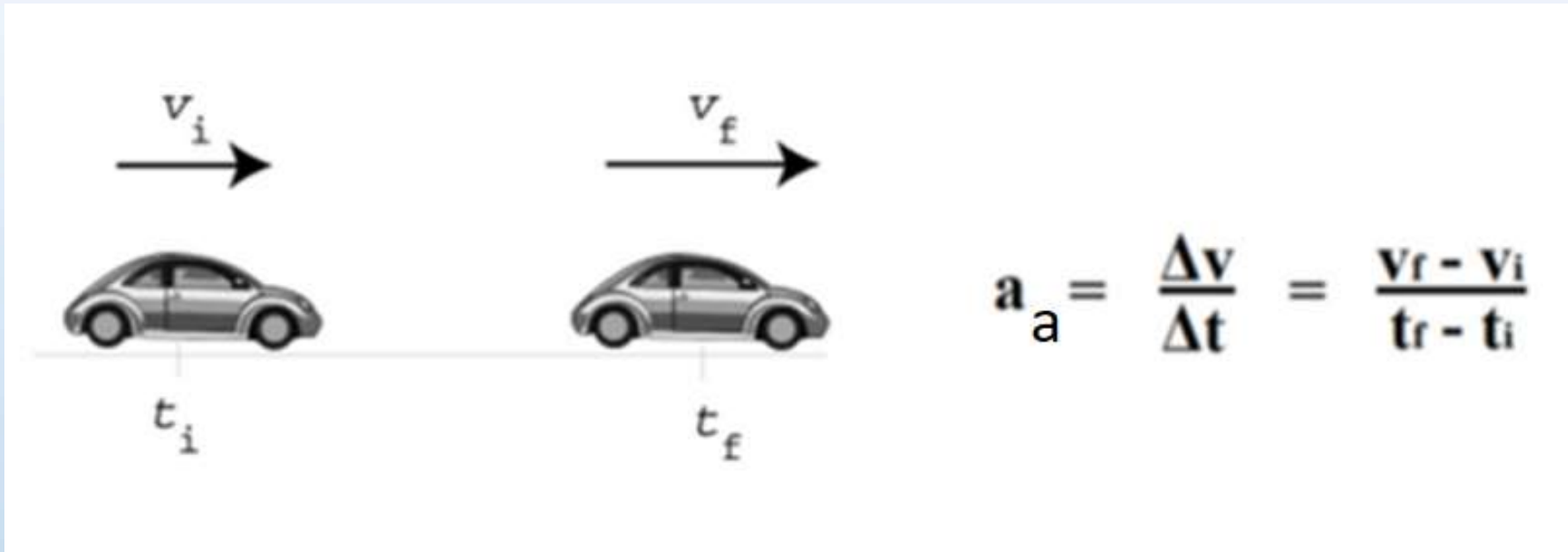


Evaluation

$$1 \text{ ms}^{-2}$$



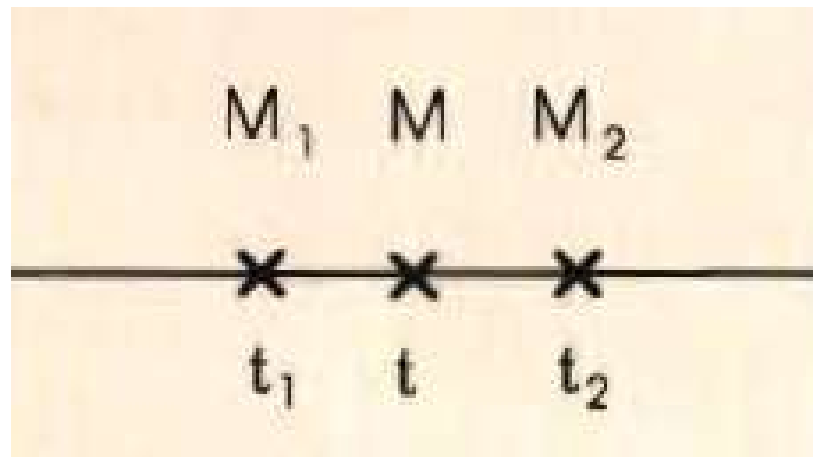
Evaluation



If the car has an initial speed of 10 m / s accelerates for 5 s to reach a final speed of 25 m / s. Calculate its average acceleration for this time interval.

(6) Instantaneous Acceleration

Similarly to instantaneous speed, the instantaneous acceleration a , at an instant t , is approximately equal to the average acceleration between two instants t_1 and t_2 very close to t so that $t_1 < t < t_2$.



Acceleration Vector: $\vec{a}$

- ❖ Origin: the position of the particle.
- ❖ Line of action: the straight line on which the particle moves.
- ❖ Direction : that of the motion if the speed increases and opposite to it if the speed decreases.
- ❖ Magnitude: the value of a .

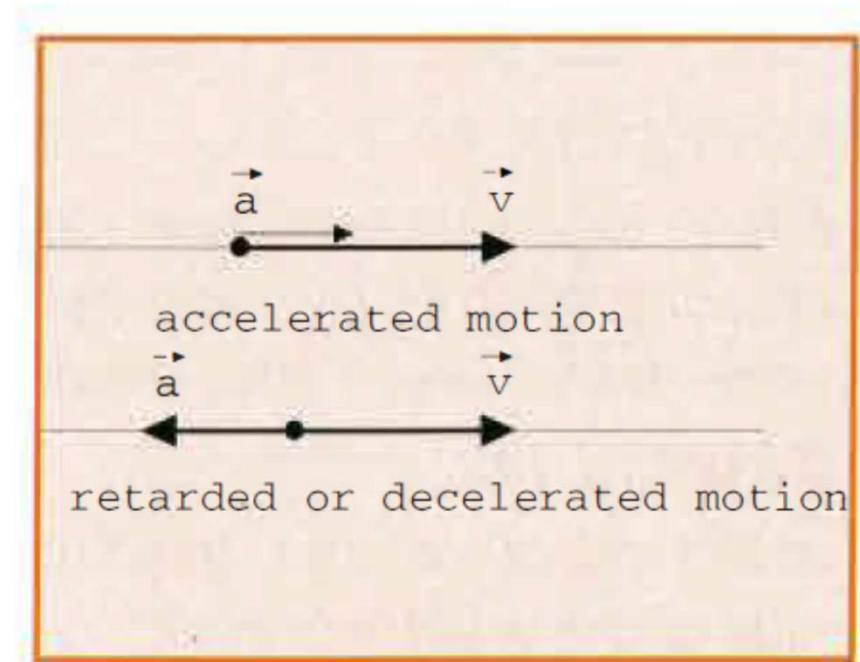


Figure 14.13. Representation of the acceleration vector.

Note

The length of the arrow that represents the acceleration vector $\vec{a}$ depends on the value of a and of course on the scale chosen.

Example 14.5

In example 14.3, determine the average acceleration of the puck between the instants t_1 and t_3 and between t_2 and t_4 (Figure 14.14).

$$V_1 = 0,108 \text{ mm / ms}$$

$$V_2 = 0,150 \text{ mm / ms}$$

$$V_3 = 0,183 \text{ mm / ms}$$

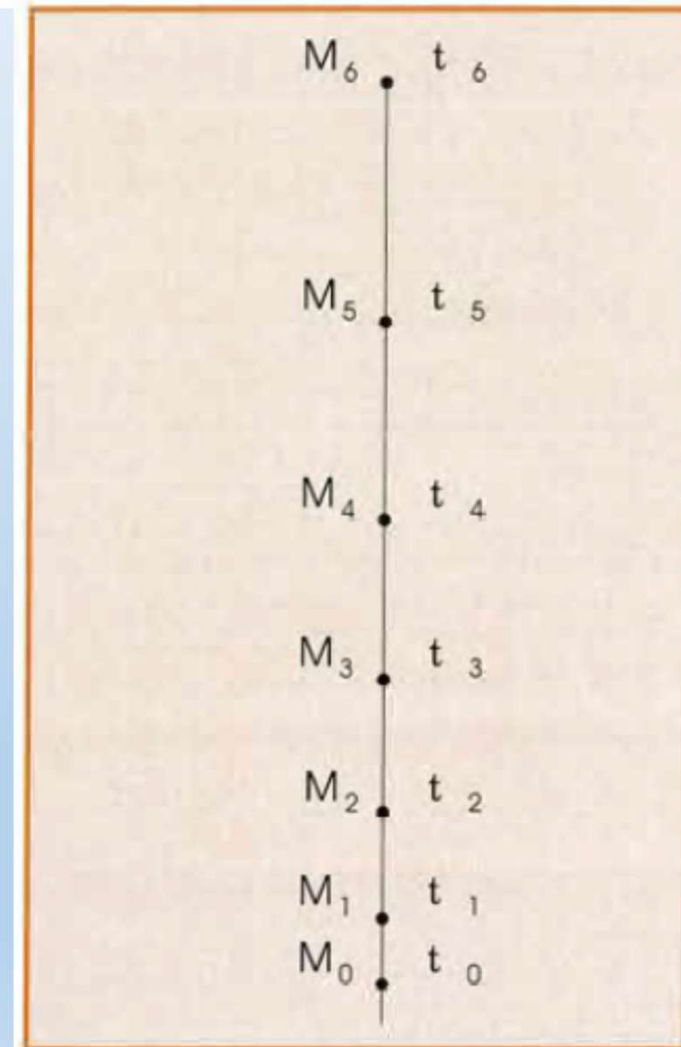


Figure 14.14. Recording of the motion of the puck.

Solution

The average accelerations are respectively:

$$a_{1,3} = (v_3 - v_1) / 2 \tau = (0.183 - 0.108) / (2 \times 60 \times 10^{-3}) \\ = 0.62 \text{ m / s}^2$$

$$a_{2,3} = (v_3 - v_2) / \tau = (0.183 - 0.150) / (60 \times 10^{-3}) \\ = 0.55 \text{ m / s}^2$$

Evaluation

Example 14.6

Calculate the average acceleration of a car moving along a rectilinear path in the positive direction, whose speed changes from 120 km/h to 130 km/h during 5 s.

Solution

The given speeds are, in m/s: ($1 \text{ km/h} = 1/3.6 \text{ m/s}$)

$$v_1 = 120 \times 10^3 / 3600 = 33.33 \text{ m/s}$$

$$v_2 = 130 \times 10^3 / 3600 = 36.11 \text{ m/s}$$

The variation of the speed is:

$$\Delta v = v_2 - v_1 = 36.11 - 33.33 = 2.78 \text{ m/s}$$

$$\Delta t = 5 \text{ s},$$

$$\text{Therefore } a_a = \Delta v / \Delta t = 2.78 / 5 = 0.56 \text{ m/s}^2$$

Summary

- * The notion of rest and motion is relative and depends on the chosen frame of reference.
- * The motion of an object is rectilinear when the trajectory is a straight line.
- * The position of a particle at a given instant is determined by the position vector.
- * In a rectilinear motion, the average speed of a particle which moves a distance d during the time interval Δt is defined by the ratio $V_a = \frac{d}{\Delta t}$
- * The instantaneous speed (at an instant t) is approximately equal to the average speed between two instants very close to t which lies between them.
- * The speed can be represented by a vector called velocity.
- * In SI units, the speed is measured in m/s or (m.s⁻¹).
- * In rectilinear motion, the average acceleration of a particle whose speed varies by Δv during the time interval Δt is defined by the ratio $\frac{\Delta v}{\Delta t}$
- * In SI units, the acceleration is measured in m/s² or (m.s⁻²).
- * The acceleration can be represented by a vector.